

Phishing Website Detection with Machine Learning: A Comparative Study of Feature Groups and Classical Classifiers

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Abstract

Phishing websites continue to be a serious cybersecurity problem because they are designed to imitate legitimate websites and trick users into giving away sensitive information such as passwords, financial data, or personal details. This project examines whether machine learning can be used to accurately detect phishing websites using structured website features, and it also looks at which categories of features are most useful for this task. The dataset used in this project contains 11,430 websites and 87 engineered features based on URL structure, hyperlink behavior, webpage signals, and trust-related information. After preprocessing the data, four machine learning models were trained and compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Among these models, Random Forest produced the strongest overall performance, with 96.02% accuracy, a 0.9603 F1 score, and a 0.9935 ROC-AUC score. Feature importance results showed that variables related to trust and reputation, such as `google_index`, `page_rank`, `web_traffic`, and `domain_age`, played a major role in the model's predictions. An ablation study also showed that using all feature groups together produced the best results, while domain and trust features were the strongest individual category. Overall, the results show that phishing websites can be detected effectively using classical machine learning methods and that combining multiple types of signals leads to the strongest performance.

1 Introduction

Phishing is a type of cyberattack in which attackers create fake websites that are meant to look legitimate in order to trick people into sharing private information. These attacks often target usernames, passwords, banking information, and other personal data. Because phishing websites are made to look convincing, they can be difficult for users to identify on their own. This has made phishing an important problem in cybersecurity and a useful area for machine learning applications.

Machine learning is especially useful in this area because it can identify patterns across large numbers of websites and use those patterns to classify whether a website is phishing or legitimate. These patterns can come from many different sources, including the structure of a URL, the behavior of hyperlinks on a page, or trust-related signals such as domain age and search indexing. While many studies have shown that phishing detection can be effective, an important question still remains: which kinds of features actually matter most?

This project examines that question by training and comparing several machine learning models on a structured phishing website dataset. More specifically, the project asks: can phishing websites be accurately detected using machine learning, and which feature groups are most useful for detection? In order to answer this, four models were compared and an ablation study was conducted to evaluate the effect of different categories of features.

This project argues that while several machine learning models perform well on phishing website classification, Random Forest provides the strongest overall performance, and the most useful results come from combining URL, hyperlink, and domain or trust-based features rather than relying on only one category alone.

2 Related Work

Previous research on phishing website detection has generally focused on using machine learning to classify suspicious websites based on structured features. A major contribution in this area comes from Hannousse and Yahiouche, who studied phishing detection using benchmark datasets and organized features into meaningful categories. Their work is especially useful for this project because it emphasizes reproducibility and helps show how different feature groups can be compared in a clear way.

Other studies have focused more directly on feature selection. For example, Chiew et al. examined how phishing detection systems can be improved by identifying which features contribute most to performance. This is important because strong classification results do not always explain why a model works well. In this project, that same concern is addressed through feature importance analysis and an ablation study.

More broadly, recent work has also explored the use of larger and more complex models for phishing detection. Aljofey et al. examined phishing detection using both URL and HTML-based features and showed that combining different signals can improve performance. While more advanced models can be effective,

they can also be harder to interpret. Because of this, classical machine learning models still provide a strong and practical way to study phishing detection, especially when the dataset already contains useful engineered features.

3 Methodology

3.1 Dataset

This project uses the Web Page Phishing Detection dataset from Kaggle and Mendeley Data. The dataset contains 11,430 website samples and 89 total columns. One column contains the raw URL, one column is the target label **status**, and the remaining 87 columns are engineered features. The dataset is balanced, with 5,715 legitimate websites and 5,715 phishing websites.

The GitHub repository for this project is available at: <https://github.com/natearteaga/phishing-website-detection-ml>

The features in the dataset can be grouped into three broad categories:

- URL and lexical features, such as **length_url**, **nb_dots**, **ratio_digits_url**, and **phish_hints**
- Hyperlink and page-based features, such as **nb_hyperlinks**, **ratio_extHyperlinks**, **safe_anchor**, and **domain_in_title**
- Domain and trust features, such as **domain_age**, **web_traffic**, **google_index**, and **page_rank**

3.2 Data Preprocessing

The dataset required only limited cleaning before modeling. There were no missing values and no duplicate rows, which made it a strong starting point for the project. The target variable **status** contained two labels: **legitimate** and **phishing**.

The preprocessing steps were as follows:

1. The raw **url** column was removed from the main modeling process because it is text-based, while the rest of the dataset already provides numeric features derived from the URL.
2. The target labels were converted into binary form, with **legitimate** = 0 and **phishing** = 1.
3. The data was split into feature variables X and target variable y .
4. The dataset was divided into training and testing sets using an 80/20 split with **random_state** = 42 and stratification to preserve the class balance.
5. Standard scaling was applied for Logistic Regression because linear models are more sensitive to differences in feature scale. The tree-based models were trained on the original numeric features.

3.3 Models

Four machine learning models were used in this project:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

These models were chosen because they provide a useful comparison between simple baseline methods and stronger ensemble methods. Logistic Regression and Decision Tree serve as baseline models, while Random Forest and Gradient Boosting allow for a more advanced comparison.

3.4 Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC. These metrics were chosen because phishing detection is a binary classification problem where both false positives and false negatives matter. Accuracy gives an overall view of performance, while precision and recall help show how well the model handles phishing predictions specifically. F1 score balances those two measures, and ROC-AUC gives a broader view of classification quality.

4 Experiments and Results

4.1 Exploratory Findings

Before training the models, exploratory data analysis was used to understand how phishing and legitimate websites differed. Several patterns appeared clearly. Phishing websites tended to have longer URLs, more suspicious symbols or structural markers, and lower domain age. Legitimate websites, on the other hand, tended to have stronger trust signals and more established domain-related characteristics.

These patterns were important because they suggested early on that the dataset contained meaningful differences that machine learning models could learn from. They also helped support the later feature importance and ablation results.

4.2 Model Comparison

The first stage of the project compared baseline and ensemble models. Table 1 shows the final results.

Table 1: Model performance comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.9361	0.9392	0.9326	0.9359	0.9814
Decision Tree	0.9243	0.9369	0.9099	0.9232	0.9624
Random Forest	0.9602	0.9566	0.9641	0.9603	0.9935
Gradient Boosting	0.9532	0.9497	0.9571	0.9534	0.9905

The results show that all four models performed well, but Random Forest produced the best overall performance. It outperformed both baseline models and slightly outperformed Gradient Boosting across the main evaluation metrics. Logistic Regression also performed strongly, which suggests that the dataset already contains a strong amount of useful signal. However, the improvement from Random Forest shows that nonlinear feature interactions also matter for phishing detection.

4.3 Feature Importance

Feature importance analysis from the Random Forest model showed that some of the most influential features were `google_index`, `page_rank`, `nb_hyperlinks`, `web_traffic`, `nb_www`, and `domain_age`. These features suggest that trust and reputation signals are especially important in distinguishing phishing websites from legitimate ones.

At the same time, other features such as `phish_hints`, `length_url`, and `ratio_digits_url` also contributed to the model. This shows that the strongest predictions come from more than one type of information. While URL-based signals matter, trust-related signals appear to matter even more in this dataset.

4.4 Ablation Study

In order to better understand which feature groups mattered most, an ablation study was performed using Random Forest across three separate feature groups and the full feature set. The results are shown in Table 2.

Table 2: Ablation study results by feature group

Feature Group	Accuracy	Precision	Recall	F1 Score	ROC-AUC
URL / Lexical Only	0.9108	0.9093	0.9125	0.9109	0.9695
Domain / Trust Only	0.9361	0.9331	0.9396	0.9364	0.9751
Hyperlink / Page Only	0.9046	0.9082	0.9003	0.9042	0.9590
All Features	0.9602	0.9566	0.9641	0.9603	0.9935

The ablation results show that using all features together produced the best performance. Among the individual feature groups, domain and trust features were the strongest on their own. URL and lexical features also performed well, while hyperlink and page-based features produced the weakest individual results, though they still remained useful. This suggests that phishing detection works best when multiple types of evidence are combined rather than treated separately.

5 Conclusion

Overall, this project shows that phishing websites can be detected effectively using classical machine learning methods. After comparing four models, Random Forest produced the strongest overall performance, with 96.02% accuracy, a 0.9603 F1 score, and a 0.9935 ROC-AUC score. These results show that machine learning can be a strong tool for classifying phishing websites when the dataset contains meaningful structured features.

The project also shows that not all feature groups contribute equally. Domain and trust features were the strongest individual category, but the best overall results came from combining all feature groups together. This suggests that phishing detection is not based on just one signal. Instead, the strongest predictions come from using URL structure, hyperlink behavior, and trust-related information together.

While the results are strong, there are still limitations. The dataset is balanced in a way that may not fully reflect real-world phishing conditions, and the models were tested on one fixed train-test split rather than repeated cross-validation. Even so, the project provides a clear and reproducible comparison of machine learning methods for phishing detection. More broadly, it shows how data-driven models can be used in cybersecurity while also raising useful questions about which kinds of information are most important in classification tasks like this one.

References

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